

Fatty liver

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1 Data sets: counting process vs. timeline

The survival package evolved along with a data set type I call a “counting process” data set. Most routines in the survival package can also deal with a “timeline” data set (hopefully all routines, soon). The `msm` and `mshmm` packages, which deal with interval censored multi-state hazard models (MSH) accept only timeline style. In a CP data set we have:

- A subject identifier
- (time1, time2] mark out a time interval
- covariates apply over the interval (time1, time2]
- likewise for stratification variables
- optionally, current state
- outcome variables, e.g., status or state, describe the change in state which occurs at time2, or censored= no change

In a timeline data set we have

- A subject identifier
- A time variable
- Covariate values that are known or measured at that time
- Missing covariates are common

The basic premise for both is a data set where each person describes a physically possible path through time. Here are three subjects from the counting process `nafls` data set:

	id	age1	age2	cstate	event	male	nafls
32	27	62.00000	62.59138	0mc	1mc	0	0
33	27	62.59138	63.68104	1mc	2mc	0	0
34	27	63.68104	77.31554	2mc	censored	0	0
120	100	50.00000	53.29911	1mc	2mc	0	1
121	100	53.29911	53.63860	2mc	3mc	0	1

122	100	53.63860	65.98084	3mc censored	0	1
159	135	40.00000	40.97194	0mc 1mc	0	1
160	135	40.97194	45.83984	1mc 2mc	0	1
161	135	45.83984	48.81588	2mc 3mc	0	1
162	135	48.81588	54.42574	3mc censored	0	1

The key criteria:

- Cannot be two places at once (no overlapping intervals)
- Cannot disappear (no holes)
- Consistent states (no jumps)
- If a state is entered, time in state must be > 0 (no teleporting)

The `survcheck` routine is very fussy, and will complain about anything that looks suspicious *on purpose*. Once you pass `survcheck`, fitting the models requires no further data manipulation. A further design goal was to make the `coxph` call for multistate hazard models (MHP) consistent for that with a simple Cox model, and the `survfit` call for multistate curves similar to the usual KM.

Last value retained (last value carried forward or last observation carried forward): If a visit has partial information, e.g., BMI is in the model and it wasn't measured on this visit, what do we use for a value? The two choices are to use the most recent available, or to extrapolate. LVCF is the default in my code, part of the data set creation (`tmerge`) for CP style data, at the analysis stage for TL style.

Immortal time bias. If any of these features of the data depend on the future, bad things happen: $x_i(t)$, $Y_i(t)$, $N_i(t)$. Run away!

2 Fatty liver disease

Non-alcoholic fatty liver disease (NAFLD) is defined by three criteria: presence of greater than 5% fat in the liver (steatosis), absence of other indications for the steatosis such as excessive alcohol consumption or certain medications, and absence of other liver disease.

MASLD is currently responsible for almost 1/3 of liver transplants and it's impact is growing, it is expected to be a major driver of hepatology practice in the coming decade, driven at least in part by the growing obesity epidemic. The `naflld` data set includes all patients with a MASLD diagnosis in Olmsted County, Minnesota between 1997 to 2014 along with up to four age and sex matched controls for each case.

The diagnostic criteria have recently been refined, along with a new label of "Metabolic dysfunction-associated fatty liver disease" (MAFLD), which explicitly recognizes the metabolic dimension.

We will model the onset of three important components of the metabolic syndrome: diabetes, hypertension, and dyslipidemia, using the model shown below. Subjects have either 0, 1, 2, or all 3 of these metabolic comorbidities.

```

> state5 <- c("0MC", "1MC", "2MC", "3MC", "death")
> tmat <- matrix(0L, 5, 5, dimnames=list(state5, state5))
> tmat[1,2] <- tmat[2,3] <- tmat[3,4] <- 1
> tmat[-5,5] <- 1
> statefig(rbind(4,1), tmat)

```

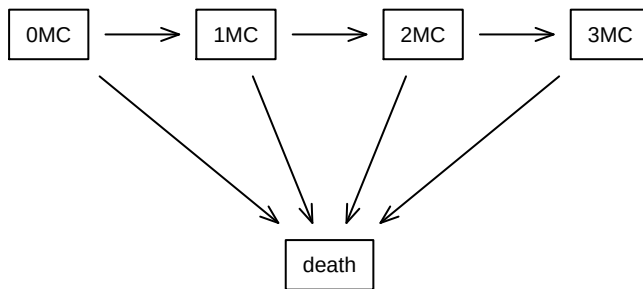


Figure 1: State space figure for NAFLD.

2.1 Data

The NAFLD data is represented as 3 data sets, `nafld1` has one observation per subject containing baseline information (age, sex, etc.), `nafld2` has information on repeated laboratory tests, e.g. blood pressure, and `nafld3` has information on yes/no endpoints. After the case-control set was assembled, we removed any subjects with less than 7 days of follow-up. These subjects add little information, and it prevents a particular confusion that can occur with a multi-day medical visit where two results from the same encounter have different dates. To protect patient confidentiality all time intervals are in days since the index date; none of the dates from the original data were retained. Subject age is their integer age at the index date, and the subject identifier is an arbitrary integer.

Here is a look at the data

```
> nafld1[1:6,]
  id age male weight height      bmi case.id futime status
1  1  57    0  60.0    163 22.69094   10630   6261      0
2  2  67    0  70.4    168 24.88403   14817    624      0
3  3  53    1 105.8    186 30.45354     3    1783      0
4  4  56    1 109.3    170 37.83010   6628   3143      0
5  5  68    1   NA     NA      NA   1871   1836      1
6  6  39    0  63.9    155 26.61559   7144   1581      0
> nafld2[1:6,]
  id days test value
1  1 -459  hdl    75
2  1 -459 chol    75
3  1  183  hdl    64
4  1  183 chol    64
5  1 2030  hdl    74
6  1 2030 chol    74
> nafld3[1:6,]
  id  days      event
1  3     0      nafld
2  4 -1287       htn
3  4 -1287 dyslipidemia
4  4 -1226    ang/isc
5  4  -90    ang/isc
6  5   617       htn
```

Start by building an analysis data set using `nafld1` and `nafld3`.

```
> ndata <- tmerge(nafld1[,1:8], nafld1, id=id, death= event(futime, status))
> ndata <- tmerge(ndata, subset(nafld3, event=="nafld"), id,
  nafld= tdc(days))
> ndata <- tmerge(ndata, subset(nafld3, event=="diabetes"), id = id,
  diabetes = tdc(days), e1= cumevent(days))
> ndata <- tmerge(ndata, subset(nafld3, event=="htn"), id = id,
```

```

      htn = tdc(days), e2 = cumevent(days))
> ndata <- tmerge(ndata, subset(nafld3, event=="dyslipidemia"), id=id,
      lipid = tdc(days), e3= cumevent(days))
> ndata <- tmerge(ndata, subset(nafld3, event %in% c("diabetes", "htn",
      "dyslipidemia")),
      id=id, comorbid= cumevent(days))
> summary(ndata)
Call:
tmerge(data1 = ndata, data2 = subset(nafld3, event %in% c("diabetes",
      "htn", "dyslipidemia")), id = id, comorbid = cumevent(days))

```

	early	late	gap	within	boundary	leading	trailing	tied	missid
death	0	0	0	0	0	0	17549	0	0
nafld	0	0	0	318	0	3533	0	0	0
diabetes	2393	0	0	1058	0	1	0	0	0
e1	2393	0	0	0	1058	1	0	0	0
htn	5022	0	0	2045	24	1	5	0	0
e2	5022	0	0	0	2069	1	5	0	0
lipid	8663	0	0	1713	82	2	2	0	0
e3	8663	0	0	0	1795	2	2	0	0
comorbid	16078	0	0	0	4922	4	7	575	0

A model for the `tmerge` function is to imagine a drawer with index cards (like an old library card catalog).

- Start with one card per subject, containing a start time, end time, covariate values, and events.
- Make additions one at a time. Each addition has an id, a time point, and a type of addition: a new variable value or new event.
 - The id must already exist.
 - If the timepoint lies within the time range of a card, replace that card with two: (old start, new time), (new time, old end). A new event is added to the first card, a new covariate value to the second. A new variable name or event is propagated to all cards for a subject.
 - If the time point aligns with the start/end of a current card or cards, update them.

The `summary` function tells us a lot about the creation process. Each addition of a new endpoint or covariate to the data generates one row in the table.

- There are 1.7549×10^4 last fu/death additions, which by definition fall at the trailing end of a subject's observation interval: they define the interval.
- There are 0 nafld splits that fall after the end of follow-up ('late'). These are subjects whose first NAFLD fell within a year of the end of their time line, and the one year delay for "confirmed" pushed them over the end. (The time value in the `nafld3` data set is 1

year after the actual notice of NAFLD; no other endpoints have this offset added). The time dependent covariate `naflld` never turns from 0 to 1 for these subjects. (Why were these subjects not removed earlier by my “at least 7 days of follow-up” rule? They are all controls for someone else and so appear in the data at a younger age than their NAFLD date.)

- 318 subjects have a NAFLD diagnosis between time 0 and last follow-up. These are subjects who were selected as matched controls for another NAFLD case at a particular age, and later were diagnosed with NAFLD themselves.
- 2393 of the diabetes diagnoses are before entry, i.e., these are the prevalent cases. One diagnosis occurred on the day of entry (“leading”), and will not be counted as a post-enrollment endpoint, all the other fall somewhere between study entry and last follow-up.
- Conversely, 5 subjects were diagnosed with hypertension at their final visit (“trailing”). These will be counted as an occurrence of a hypertension event (`e2`), but the time dependent covariate `htn` will never become 1.
- 575 of the total comorbidity counts are tied. These are subjects for whom the first diagnosis of 2 of the 3 conditions happened on the same office visit, the cumulative count will jump by 2. (We will see below that 4 subjects had all 3 on the same day.) Many times ties indicate a data error.

Such a detailed look at data set construction may seem over zealous. Our experience is that issues with covariate and event timing occur in nearly all data sets, large or small. The 13 NAFLD cases “after last follow-up” were for instance both a surprise and a puzzle to us; but we have learned through experience that it is best not to proceed until such puzzles are understood. (This particular one was benign.) If, for instance, some condition is noted at autopsy, do we want the related time dependent covariate to change before or after the death event? Some sort of decision has to be made, and it is better to look and understand than to blindly accept an arbitrary programming default.

Create the covariates for current state and the analysis endpoint. It is important that data manipulations like this occur *after* the final `tmerge` call. Successive `tmerge` calls keep track of the time scale, time-dependent and event covariates, passing the information forward from call to call, but this information is lost when the resulting data frame is manipulated. (The loss is intentional: we won’t know if one of the tracked variables has changed.)

The `tmerge` call above used the `cumevent` verb to count comorbidities, and the first line below verifies that no subject had diabetes, for instance, coded more than once. For this analysis we think of the three conditions as one-time outcomes, you can’t get diabetes twice. When the outcome data set is the result of electronic capture one could easily have a diabetes code at every visit, in which case the cumulative count of all events would not be the total number of distinct comorbidities. In this particular data set the diabetes codes had already been preprocessed so that the data set contains only the first diabetes diagnosis, and likewise with hypertension and dyslipidemia. (In counterpoint, the `nafld3` data set has multiple myocardial infarctions for some subjects, since MI can happen more than once.)

```

> with(ndata, if (any(e1>1 | e2>1 | e3>1)) stop("multiple events"))
> ndata$cstate <- with(ndata, factor(diabetes + htn + lipid, 0:3,
                                     c("0mc", "1mc", "2mc", "3mc")))
> temp <- with(ndata, ifelse(death, 4, comorbid))
> ndata$event <- factor(temp, 0:4,
                        c("censored", "1mc", "2mc", "3mc", "death"))
> ndata$age1 <- ndata$age + ndata$tstart/365.25 # analysis on age scale
> ndata$age2 <- ndata$age + ndata$tstop/365.25
>
> check1 <- survcheck(Surv(age1, age2, event) ~ 1, data=ndata,
                      id=id, istate=cstate)
> check1
Call:
survcheck(formula = Surv(age1, age2, event) ~ 1, data = ndata,
          id = id, istate = cstate)

```

Unique identifiers	Observations	Transitions
17549	22683	6186

Transitions table:

	to				
from	1mc	2mc	3mc	death	(censored)
0mc	1829	70	4	263	5705
1mc	0	1843	28	243	4567
2mc	0	0	1048	417	3687
3mc	0	0	0	441	2220
death	0	0	0	0	0

Number of subjects with 0, 1, ... transitions to each state:

	count				
state	0	1	2	3	4
1mc	15720	1829	0	0	0
2mc	15636	1913	0	0	0
3mc	16469	1080	0	0	0
death	16185	1364	0	0	0
(any)	12733	3673	938	183	22

This is a rich data set with a large number of transitions: over 1/4 of the participants have at least one event, and there are 22 subjects who transition through all 5 possible states (4 transitions). Subjects do not all enter the study in the same state; about 14% have diabetes at the time of recruitment, for instance. One major difference between current state and outcome is that the current state endures across intervals: it is based on `tdc` variables while the outcome is based on `event` operators. If a subject has time-dependent covariates, there may be intermediate intervals where a covariate changed but an outcome did not occur; current state will endure across intervals but the intermediate outcome will be “censor”.

```

> states <- c("No comorbidity", "1 comorbidity", "2 comorbidities",
             "3 comorbidities", "Death")
> cmat <- matrix(0, 5,5)
> cmat[,5] <- 1
> cmat[1,2] <- cmat[2,3] <- cmat[3,4] <- 1
> cmat[1,3] <- cmat[2,4] <- 1.6
> cmat[1,4] <- 1.6
> dimnames(cmat) <- list(states, states)
> statefig(cbind(4,1), cmat, alty=c(1,2,1,2,2,1,1,1,1,1))

```

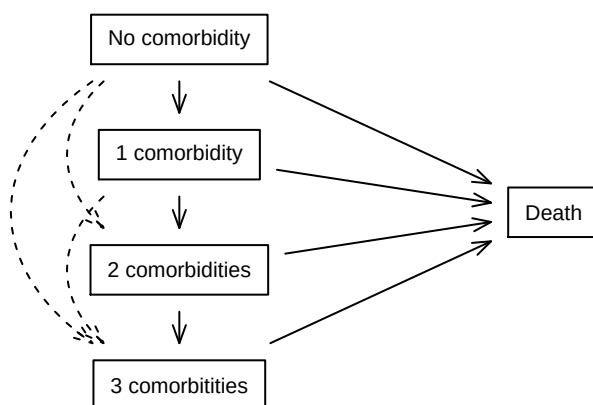


Figure 2: Augmented state space figure.

We see a number of subjects who “jump” states, e.g., directly from 0 to 2 comorbidities. This serves to remind us that this is actually a model of time until *detected* comorbidity; which will often have such jumps even if the underlying biology is continuous. The data look like the figure 2, where the dotted lines are transformations that we observe, but would not be present if the subjects were monitored continuously. A call to the **survcheck** routine is almost mandatory for a complex setup like this, to ensure that the data set which has been built is what you intended to build.

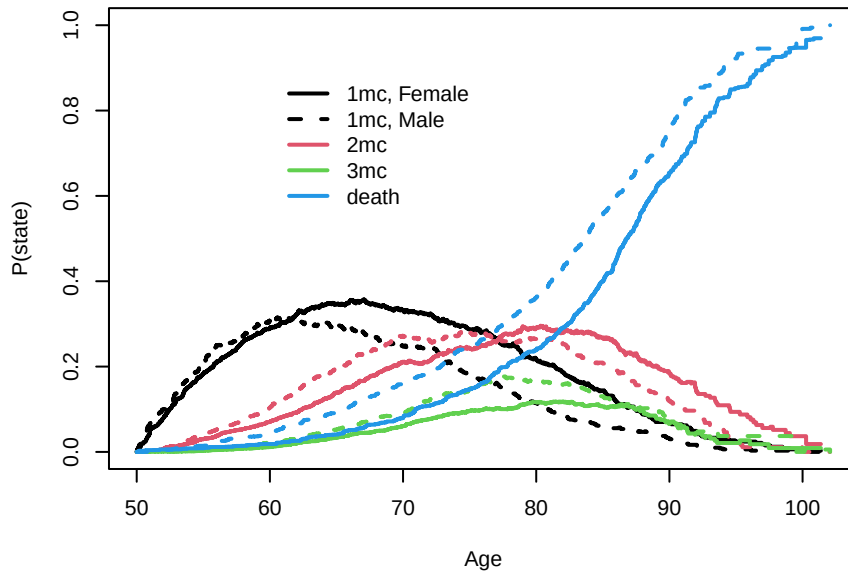
Calling **survcheck** with `~1` on the right hand side or with the covariates for the model on the right hand side will potentially give different event counts, due to the removal of rows with a missing value. Both can be useful summaries. For a multi-state coxph model neither may be exactly correct, however. If the model contains a covariate which applies only to certain transitions, then events that do not depend on that covariate will be retained, while event occurrences that do depend on the covariate will be dropped, leading to counts that may be intermediate between the two **survcheck** outputs.

3 Fitting the model

3.1 Aalen-Johansen

First, look at an AJ estimate on age scale.

```
> aj1 <- survfit(Surv(age1, age2, event) ~ male, ndata, id=id,
                 istate=cstate, start.time=50, p0=c(1,0,0,0,0))
> dim(aj1)
strata states
      2      5
> plot(aj1[, -1], lty=1:2, col=rep(1:4, each=2), lwd=2,
       xlab="Age", ylab= "P(state)")
> legend(60, .9, c("1mc, Female", "1mc, Male", "2mc", "3mc", "death"),
       lty=c(1,2,1,1,1), lwd=2, col=c(1,1,2,3,4), bty='n')
```



At each event time t the AJ algorithm makes a matrix

$$A(t) = \begin{pmatrix} \square & \lambda_{12}(t) & \lambda_{13}(t) & \dots \\ \lambda_{21}(t) & \square & \lambda_{23}(t) & \dots \\ \lambda_{31}(t) & \lambda_{32} & \square & \dots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

$$H(t) = I + A$$

$$\lambda_{jk} = \frac{\sum_i dN_{ijk}(t)}{\sum_i Y_{ij}(t)}$$

The diagonal of the A matrix is such that each row sums to 0, for H each row sums to 1. Then

$$p(t) = p(t_0) \prod_{t_0 \leq s \leq t} H(s)$$

is the Aalen-Johansen. $p_j(t) = P(\text{in state } j \text{ at time } t)$, $0 \leq p_j(t) \leq 1$, $\sum_j p_j(t) = 1$.

- You can use any starting time t_0 that you like
- You can use any starting distribution you like
- survfit default
 - t_0 = common starting time, if there is one, else the first transition time
 - $p(t_0)$ = empirical distribution at t_0 or $t_0 -$
- When using age scale the default is rarely useful

4 Multistate hazard model

Since age is the dominant driver of the transitions we have chosen to do the fits directly on age scale rather than model the age effect. We force common coefficients for the transitions from 0 comorbidities to 1, 2 or 3, and for transitions from 1 comorbidity to 2 or 3. This is essentially a model of “any progression” from a given state. We also force the effect of male sex to be the same for any transition to death.

```
> nfit1 <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                    1:c(2,3,4) ~ nafld+ male/ common,
                    2:c(3,4) ~ nafld+ male/common,
                    0:"death" ~ male / common),
                data=ndata, id=id, istate=cstate)
> print(nfit1, digits=1)
Call:
coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
                    1:c(2, 3, 4) ~ nafld + male/common, 2:c(3, 4) ~ nafld + male/common,
                    0:"death" ~ male/common), data = ndata, id = id, istate = cstate)
```

```
1:2, 1:3, 1:4 coef exp(coef) robust se z      p
nafld 0.91      2.50      0.07 14 <2e-16
male  0.18      1.20      0.05  4  2e-04
```

```
2:3, 2:4 coef exp(coef) robust se z      p
nafld 0.52      1.68      0.06  9 <2e-16
male  0.25      1.28      0.05  5  9e-07
```

```

3:4      coef exp(coef) robust se z      p
nafld 0.48      1.62      0.07 7 9e-13
male 0.15      1.16      0.07 2 0.02

```

```

1:5      coef exp(coef) robust se z      p
nafld 0.63      1.88      0.23 3 0.006
male 0.33      1.39      0.05 6 2e-09

```

```

2:5      coef exp(coef) robust se z      p
nafld 0.53      1.71      0.15 4 4e-04
male 0.33      1.39      0.05 6 2e-09

```

```

3:5      coef exp(coef) robust se z      p
nafld 0.55      1.74      0.11 5 3e-07
male 0.33      1.39      0.05 6 2e-09

```

```

4:5      coef exp(coef) robust se z      p
nafld 0.07      1.07      0.10 0.7 0.5
male 0.33      1.39      0.05 6.0 2e-09

```

States: 1= 0mc, 2= 1mc, 3= 2mc, 4= 3mc, 5= death

Likelihood ratio test=448 on 11 df, p=<2e-16
n= 22683, unique id= 17549, number of events= 6186

A list has been used as the formula for the `coxph` call. The first element is a standard formula, and will be the default for all of the transitions found in the model. Further elements of the list are pseudo formulas, which specify a set of states on the left and covariates on the right, along with the optional modifier `/common`. There are multiple ways to specify a set of transitions either by name or by number, the value 0 is shorthand for “any state”. The coefficient matrix reveals that the 1:2, 1:3, and 1:4 transitions all share the same coefficients, as intended.

Here is a non-parametric version

Where do the standard errors come from?

```

> ijmat <- resid(aj2, type="sojourn", time=c(50, 100))
> dim(ndata)
[1] 22683    23

```

	NAFLD		Male	
	HR	p	HR	p
0:(1-3)	2.50	< 0.001	1.20	< 0.001
1:(2-3)	1.68	< 0.001	1.28	< 0.001
2:3	1.62	< 0.001	1.16	0.0220
0:death	1.88	0.0057	1.39	< 0.001
1:death	1.71	< 0.001	1.39	< 0.001
2:death	1.74	< 0.001	1.39	< 0.001
3:death	1.07	0.4659	1.39	< 0.001

Table 1: Estimated hazard ratio and p-values for the multistate model.

```

> dummy <- expand.grid(naflld=0:1, male=0:1)
> nsurv <- survfit(nfit1, newdata=dummy, start.time=50, p0=c(1,0,0,0,0))
> temp <- summary(nsurv, rmean=100)$table
> rmat <- matrix(temp[, 'rmean'], ncol=4, byrow=TRUE)
> colnames(rmat) <- c("Female, Normal", "Female, MAFLD", "Male, Normal",
  "Male, MAFLD")
> oldpar <- par(mar=c(5,9,1,1))
> barplot(rmat[, c(2,1,3,4)], horiz=TRUE, xlim=c(0,50),
  col=c("green","orange", "red", "darkred", "white"), las=1,
  xlab="Years in state, from age 50")

```

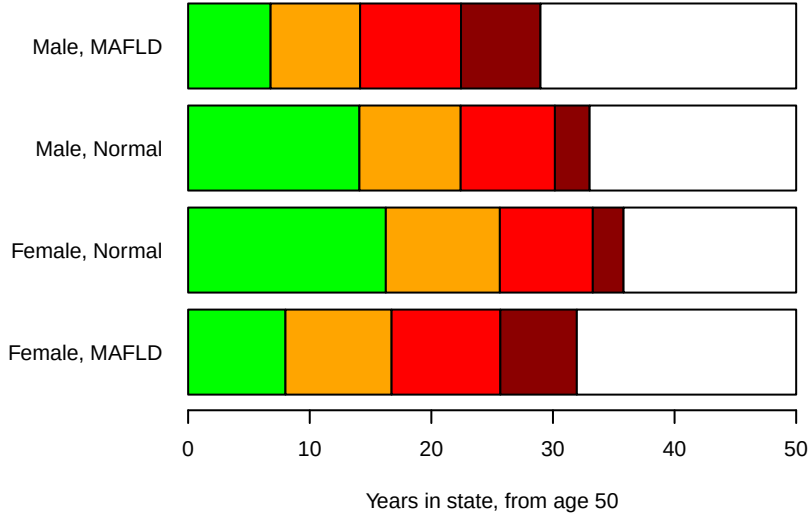


Figure 3: Predicted mean time in state for the MAFLD data. From left to right are 0, 1, 2, 3 cormorbidities, and death (white).

```

> aj2 <- survfit(Surv(age1, age2, event) ~ male+ nafld, ndata, id=id,
                 istate=cstate, start.time=50, p0=c(1,0,0,0,0))
> dim(aj2)
strata states
      4      5
> temp2 <- summary(aj2, rmean=100)$table
> round(temp2[1:7,], 2)
               n nevent rmean se(rmean)
male=0, nafld=0, 0mc 7153      0 16.28      0.45
male=0, nafld=1, 0mc 1973      0  7.99      0.93
male=1, nafld=0, 0mc 5883      0 13.84      0.48
male=1, nafld=1, 0mc 1644      0  7.99      0.81
male=0, nafld=0, 1mc 7153     552  9.79      0.36
male=0, nafld=1, 1mc 1973      78  9.14      0.64
male=1, nafld=0, 1mc 5883     420  7.62      0.33
> rmat <- matrix(temp2[, 'rmean'], ncol=4, byrow=TRUE)
> colnames(rmat) <- c("Female, Normal", "Female, MAFLD", "Male, Normal",
                     "Male, MAFLD")
> oldpar <- par(mar=c(5,9,1,1))
> barplot(rmat[, c(2,1,3,4)], horiz=TRUE, xlim=c(0,50),
          col=c("green","orange", "red", "darkred", "white"), las=1,
          xlab="Years in state, from age 50")

```

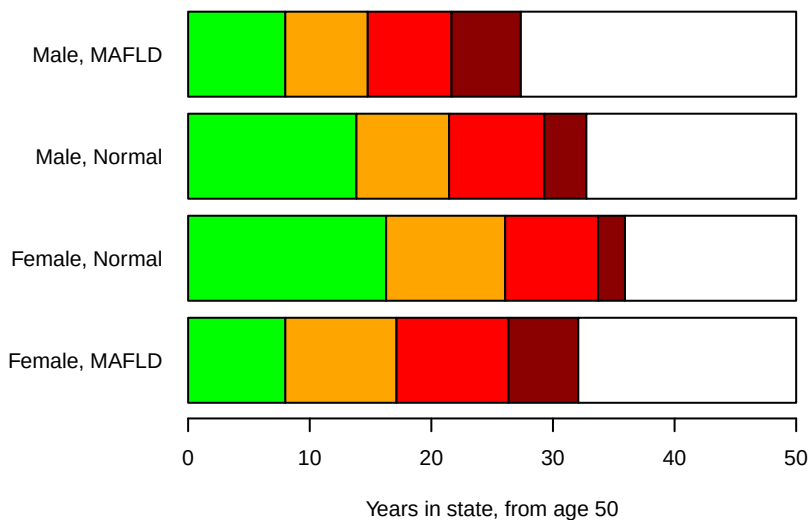


Figure 4: Non parametric estimate using the AJ